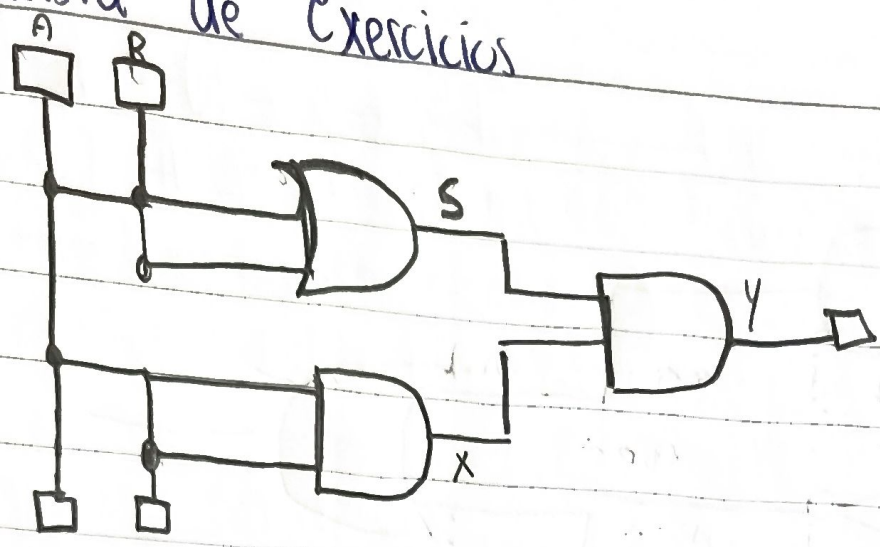


Lista de Exercícios

①



$$Y = S \cdot X$$

$$Y = (A+B) \cdot (A \cdot B)$$

$$Y = AAB + ABB$$

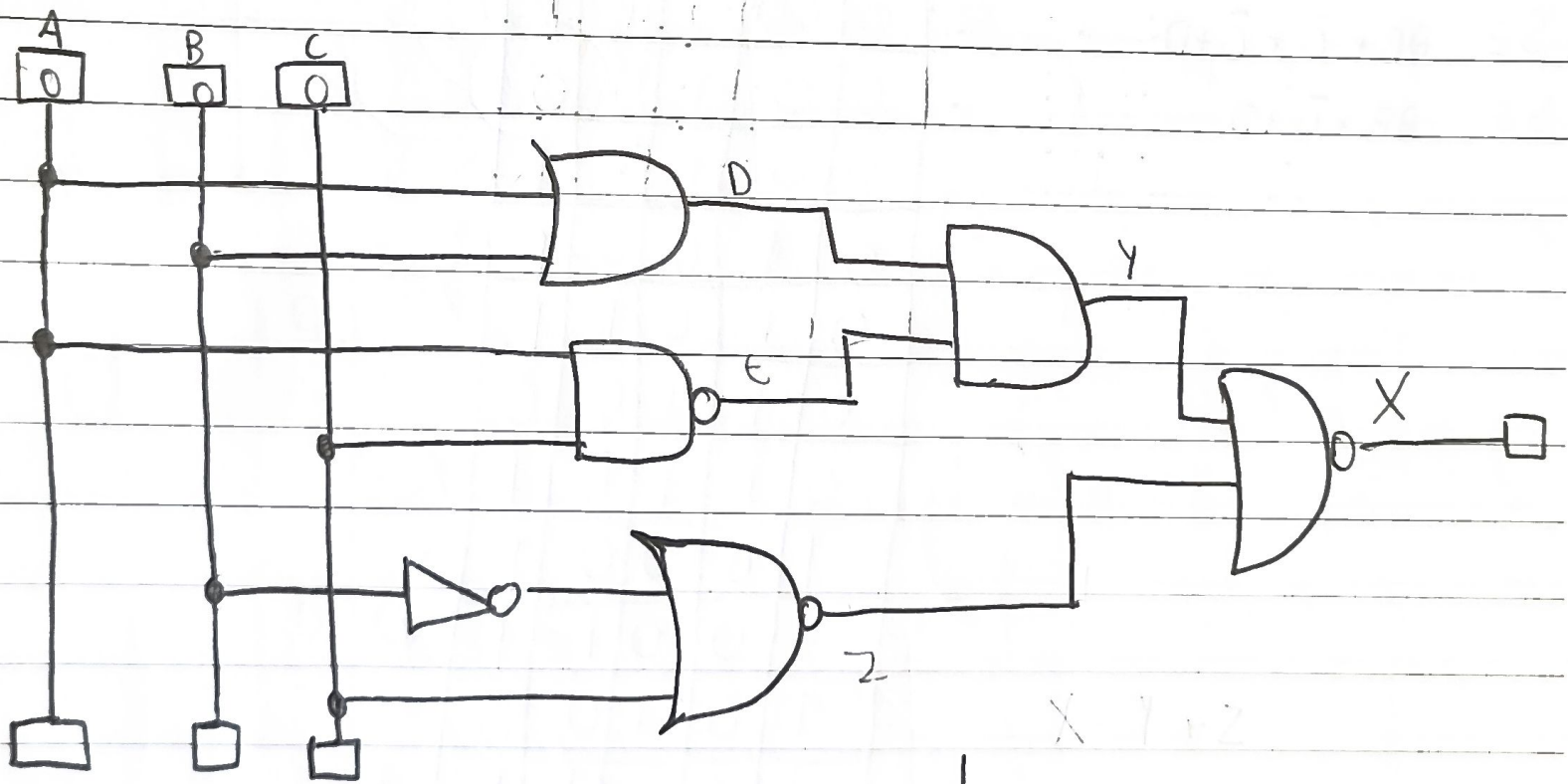
$$AB + AB \quad A \cdot B$$

$$A \cdot B$$

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

ab

b



A	B	C	S
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$X(A+B) \cdot (\overline{AC}) + (\overline{B}+C)$$

$$X \cdot Y \cdot Z \cdot 1$$

$$(A+B) \cdot (\overline{A}+\overline{C}) + (B \cdot \overline{C})$$

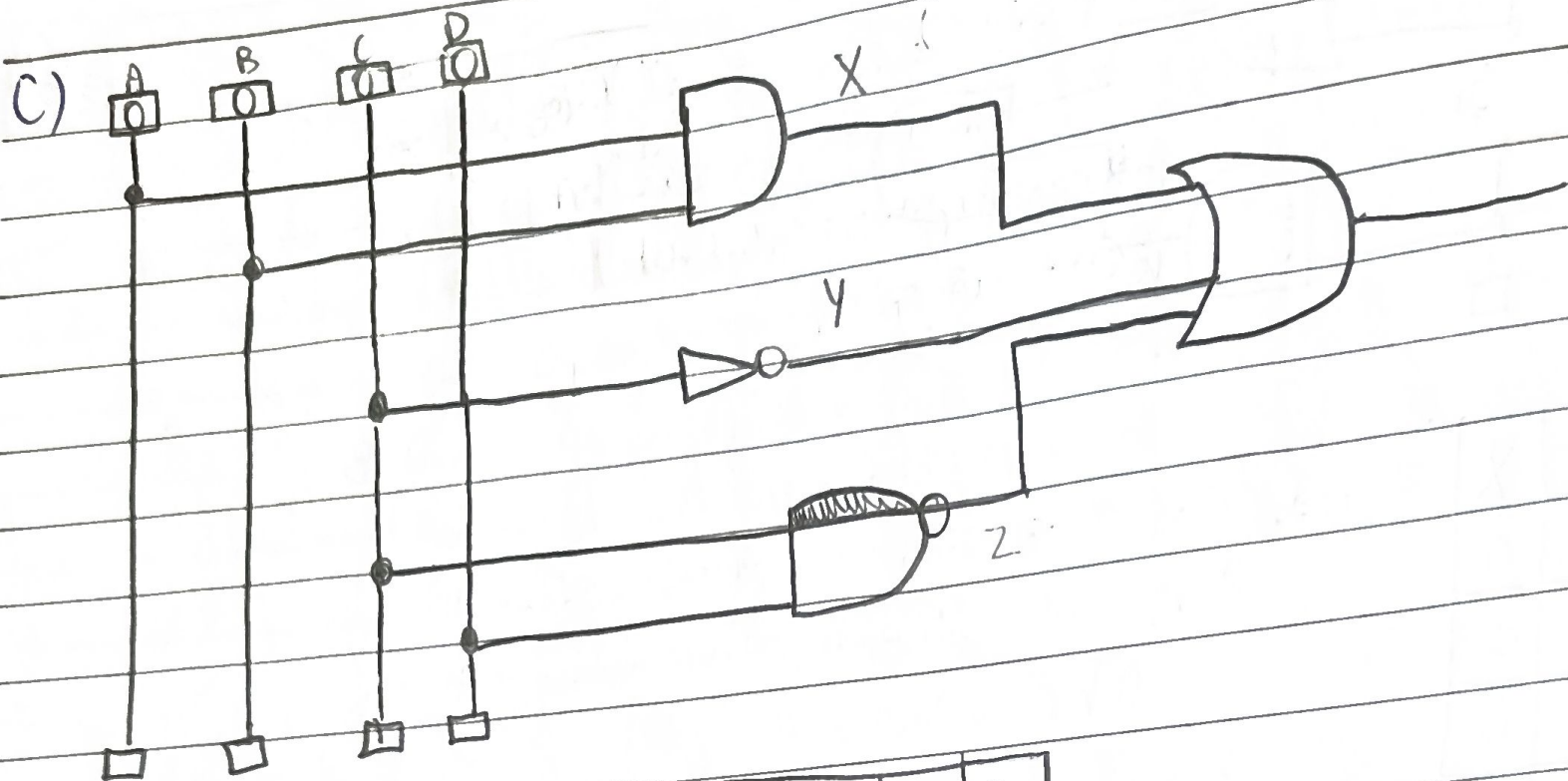
$$(\overline{A}\overline{B}) + (A \cdot C) + (\overline{B}+C)$$

$$\overline{A}\overline{B}\overline{B} + \overline{A}\overline{B}C + A\overline{C}\overline{B} + A\overline{C}C$$

$$\overline{A}\overline{B} + \overline{A}\overline{B}C + A\overline{C}\overline{B} + A\overline{C}$$

$$\overline{A}\overline{B} + A\overline{C}(\overline{B}+B)$$

$$\overline{a}\overline{b}\overline{c} + \overline{a}\overline{b}c + a\overline{b}\overline{c} + abc$$



$$S \equiv X + Y + Z$$

$$S \equiv AB + \bar{C} + \bar{C}D$$

$$S \equiv AB + \bar{C} + \bar{C} + D$$

$$S \equiv AB + \bar{C} + D$$

A	B	C	D	S	
0	0	0	0	1	X
0	0	0	1	1	X
0	0	1	0	0	
0	0	1	1	1	X
0	1	0	0	1	X
0	1	0	1	1	X
0	1	1	0	0	
0	1	1	1	1	X
1	0	0	0	1	X
1	0	0	1	1	X
1	0	1	0	0	
1	0	1	1	1	X
1	1	0	0	1	X
1	1	0	1	1	X
1	1	1	0	1	
1	1	1	1	1	

$$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD + AB\bar{C}\bar{D} + AB\bar{C}D + ABC\bar{D} + ABCD$$

Circuito (d)

$$(B \oplus D) + ((\overline{A+B+C} + \overline{A \cdot C \cdot D}) \cdot (C)) + \overline{A+B+C} \cdot \overline{D}$$

$$(\overline{B}D + B\overline{D}) + ((\overline{A+B+C} \cdot A \cdot C \cdot D) \cdot (C)) + (\overline{A+B+C} \cdot \overline{D})$$

$$(\overline{B}D + B\overline{D}) + ((\overline{A}A \cdot C \cdot D + A \cdot B \cdot C \cdot D + A \cdot \overline{B} \cdot C \cdot D) \cdot (C)) + (\overline{A+B+C} \cdot \overline{D})$$

$$(\overline{B}D + B\overline{D}) + (A \cdot B \cdot C \cdot C \cdot D + A \cdot \overline{B} \cdot C \cdot D) + (\overline{A+B+C} \cdot \overline{D})$$

$$S \equiv \overline{B}D + \overline{A+B+C} \cdot \overline{D}$$

A	B	C	D	S	
0	0	0	0	1	$\overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D} + \overline{A} \cdot \overline{B} \cdot C \cdot \overline{D} + \overline{A} \cdot B \cdot \overline{C} \cdot \overline{D} + \overline{A} \cdot B \cdot C \cdot \overline{D} + \overline{A} \cdot \overline{C} \cdot \overline{D} + \overline{A} \cdot B \cdot \overline{C} \cdot D + \overline{A} \cdot B \cdot C \cdot D + \overline{A} \cdot C \cdot D$
0	0	0	1	1	$A \cdot \overline{B} \cdot \overline{C} \cdot \overline{D} + A \cdot \overline{B} \cdot C \cdot \overline{D} + A \cdot B \cdot \overline{C} \cdot \overline{D} + A \cdot B \cdot C \cdot \overline{D} + A \cdot \overline{C} \cdot \overline{D} + A \cdot B \cdot \overline{C} \cdot D + A \cdot B \cdot C \cdot D + A \cdot C \cdot D$
0	0	1	0	1	x
0	0	1	1	1	x
0	1	0	0	1	x
0	1	0	1	1	x
0	1	1	0	1	x
0	1	1	1	1	x
1	0	0	0	1	
1	0	0	1	1	
1	0	1	0	1	
1	0	1	1	1	
1	1	0	0	1	
1	1	0	1	1	
1	1	1	1	1	

Circuito e)

$$\overline{B \cdot \bar{D} + A} \cdot \overline{\bar{B}D + (D \cdot C + ((\bar{A} + C) \cdot (\overline{B \cdot D}))}$$

$$(\overline{B \cdot \bar{D} + A}) + (\overline{B + \bar{D}}) \cdot (\overline{\bar{C} + \bar{D}}) \cdot \bar{C} \cdot ((A \cdot \bar{C}) + (B \cdot D))$$

$$(\overline{AB + A\bar{D}}) + (\overline{B\bar{C} + B\bar{D} + \bar{C}\bar{D} + \bar{D}\bar{D}}) \cdot A\bar{C}\bar{C} + B \cdot D$$

$$A\bar{C} + B$$

$$AB + A\bar{D} + B\bar{C} + B\bar{D} + \bar{C}\bar{D} + \bar{D} \cdot (A\bar{C} + BD\bar{C})$$

$$(AB + B\bar{C} + \bar{D}) \cdot (A\bar{C} + BD\bar{C})$$

$$(AB + B\bar{C} + \bar{D}) \cdot \bar{C} \cdot (A + BD)$$

$$(AB\bar{C} + B\bar{C}\bar{C} + \bar{D}\bar{C}) \cdot A + BD$$

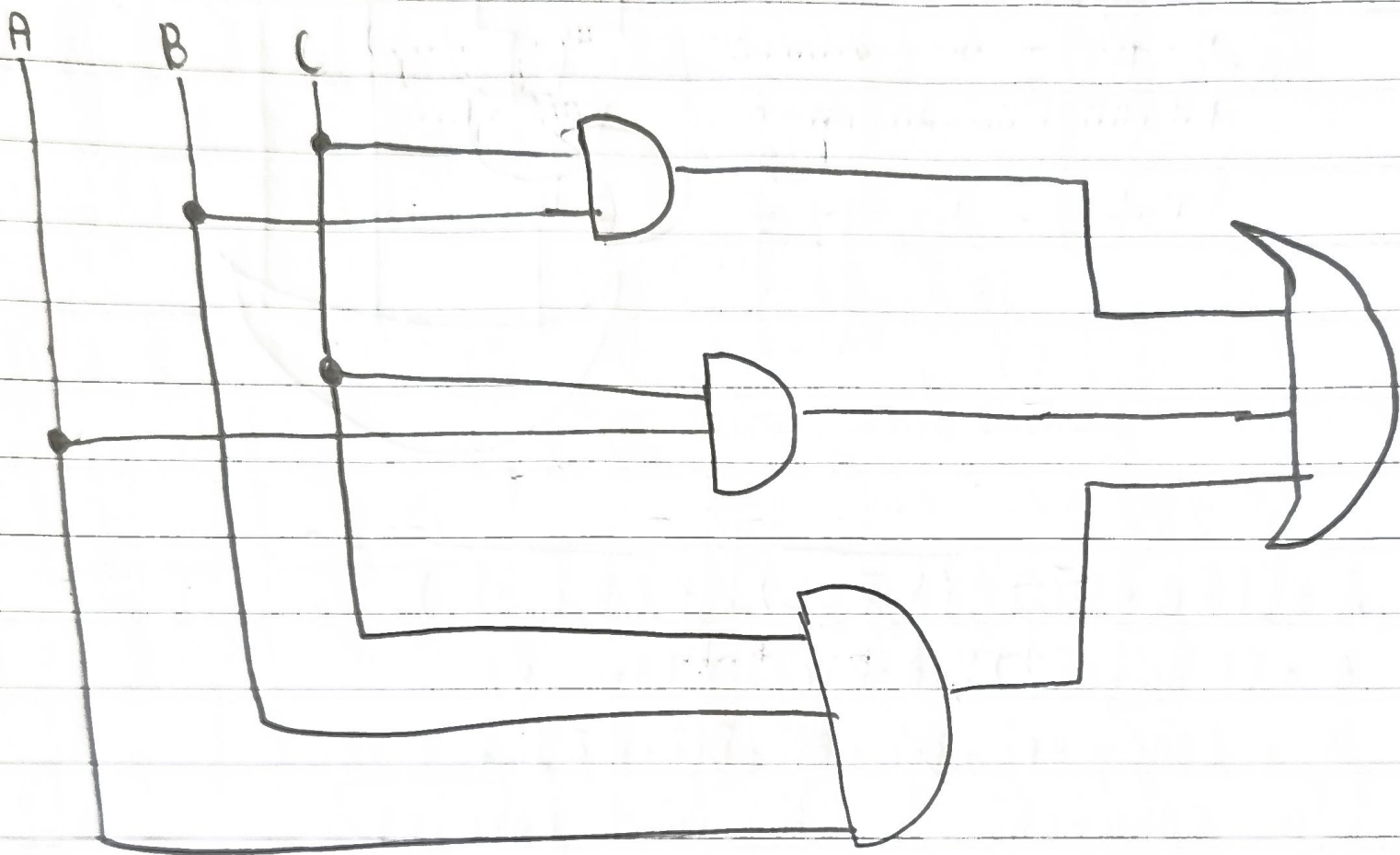
$$AAB\bar{C} + AB\bar{C}\bar{C} + A\bar{D}\bar{C} + BD$$

$$AB\bar{C} + A\bar{D}\bar{C} + A\bar{D}\bar{C} + BD$$

$$\bar{C} \cdot (AB + AD)$$

$$AB\bar{C} + A\bar{C} + BD$$

2 a) $ABC + [(A+B).C]$
 $ABC + AC + BC$

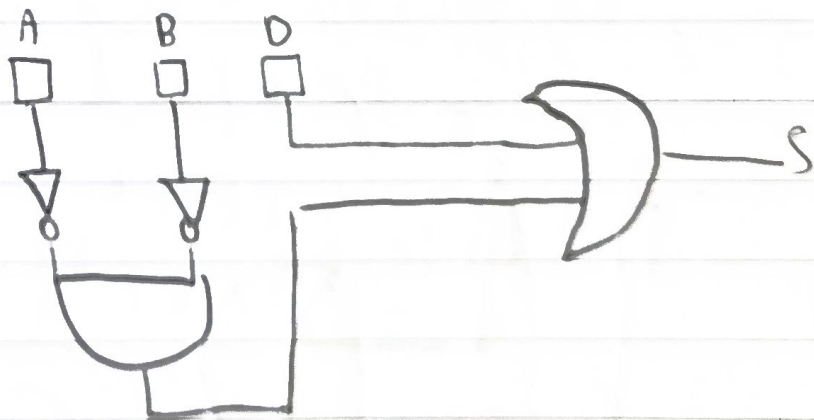


b) $(\bar{A} \cdot \bar{B}) + (\bar{C} + \bar{D}) \cdot \bar{D}$

$\bar{A} \cdot \bar{B} + \bar{C} \bar{D} + \bar{D} \bar{D}$

$\bar{A} \bar{B} + \bar{C} \bar{D} + \bar{D}$

$\bar{A} \bar{B} + \bar{D}$



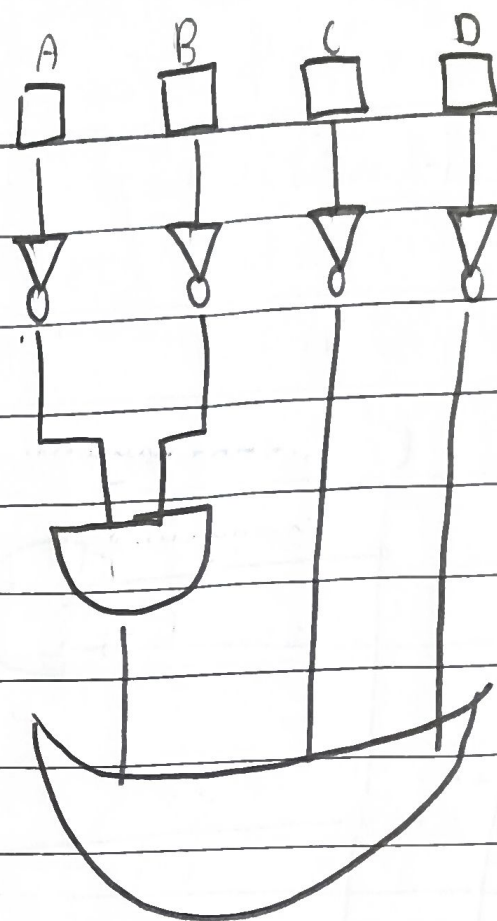


$$c) \overline{(A+B)} \cdot C + [D \cdot \overline{(B+C)}]$$

$$(\bar{A} \cdot \bar{B}) + \bar{C} + \bar{D} + \bar{B} \cdot \bar{C}$$

$$\bar{A} \bar{B} + \bar{C} + \bar{D} + \bar{B} \bar{C}$$

$$\bar{A} \bar{B} + \bar{C} + \bar{D}$$



$$d) A + [(\bar{B}C + B\bar{C}) \cdot \overline{(\bar{A} \cdot \bar{B} \cdot C)}] + (\bar{A} \cdot \bar{C} + \bar{B})$$

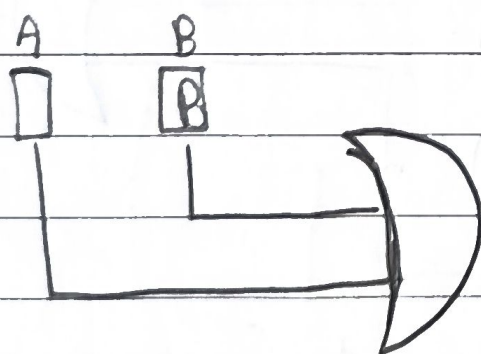
$$A + [(\bar{B}C + B\bar{C}) \cdot (A+B+\bar{C})] + (A+C) \cdot B$$

$$A + [A\bar{B}C + A\bar{B}\bar{C} + \bar{B}BC + B\bar{B}C + \bar{B}\bar{C}C + B\bar{C}\bar{C}] + BA + BC$$

$$A + [A\bar{B}C + A\bar{B}\bar{C} + \bar{B}C + B\bar{C}] + (A+C) \cdot B$$

$$A + A\bar{B}C + A\bar{B}\bar{C} + \bar{B}C + B\bar{C} + \cancel{A}B + BC$$

$$A + B$$



e) $\bar{C} \cdot [A \cdot \bar{B} + B \cdot (\bar{A} + C)]$

$\bar{C} \cdot [(\bar{A} + B) \cdot \bar{B} + (A \cdot \bar{C})]$

$\bar{C} \cdot [(\bar{A} + B) \cdot (A \cdot \bar{B}) \cdot (\bar{B} + \bar{C})]$

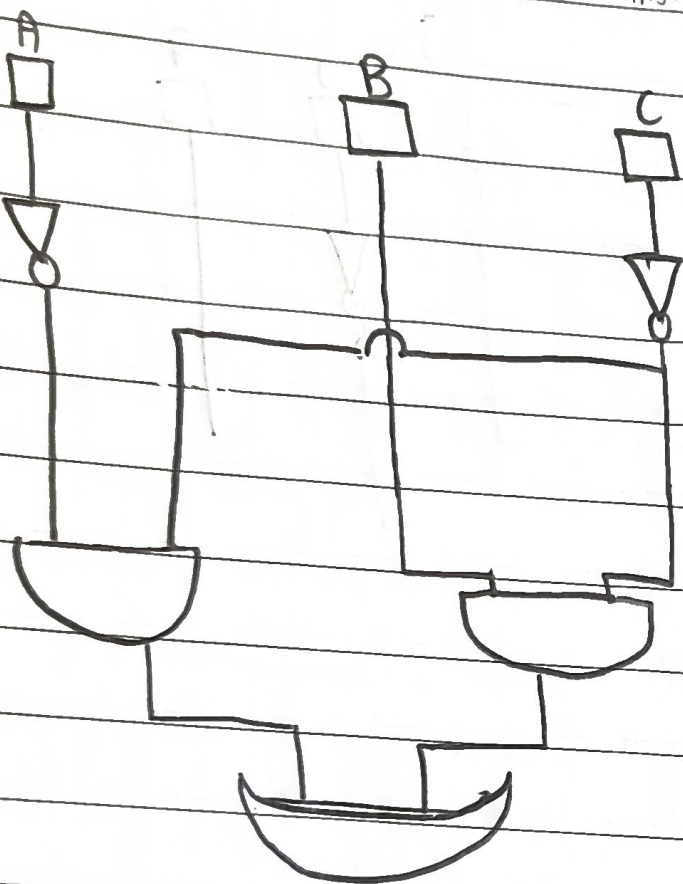
$\bar{A}A + \bar{A}\bar{B} + \bar{A}\bar{B} \cdot \bar{A}\bar{C} + AB + B\bar{B} + B\bar{B} + B\bar{C}$

$\bar{C} \cdot (\bar{A}\bar{B} + \bar{A}\bar{C} + AB + B\bar{C})$

$\bar{A}\bar{B}\bar{C} + \bar{A}\bar{C}\bar{C} + AB\bar{C} + B\bar{C}\bar{C}$

$\rightarrow \bar{A}\bar{B}\bar{C} + \bar{A}\bar{C} + AB\bar{C} + B\bar{C}$

$\bar{A}\bar{C} + B\bar{C}$



f) $ABC + A\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + C$

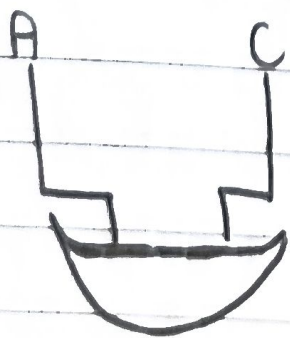
$+ \bar{A}B\bar{C} + A\bar{B}\bar{C} + C$

$ABC + A\bar{B}\bar{C} + C$

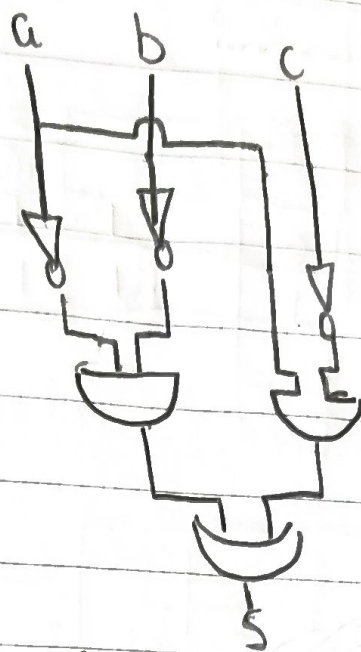
$A\bar{C} + C$

$\rightarrow$

$A + C \cdot C + \bar{C} \rightarrow A + C$



5) $\bar{a}\bar{b}\bar{c} + \bar{a}bc + a\bar{b}\bar{c} + ab\bar{c}$
 $\bar{a}b + a\bar{c}$



b) $\bar{a}\bar{b}\bar{c}\bar{d} + \bar{a}b\bar{c}\bar{d} + a\bar{b}\bar{c}\bar{d} + \bar{a}\bar{b}c\bar{d} + \bar{a}b\bar{c}d + a\bar{b}cd + ab\bar{c}d + abcd$
 $\bar{a}b\bar{d} + \bar{a}b\bar{d} + a\bar{c}d + abc$
 $\bar{a}b\bar{d} + \bar{a}b\bar{d} + a\bar{c}d + abc = \bar{a}b\bar{d} + a\bar{c}d + abc = ac(b+d)$

ab \ cd	00	01	11	10
00		✓		✓
01				
11			✓	✓
10		✓	✓	✓

$\bar{a}b\bar{d} + \bar{a}b\bar{d} + a\bar{c}d + abc$

$\bar{d}(\bar{b} + a)$

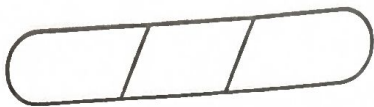
$\bar{a}\bar{b}c\bar{d} + \bar{a}b\bar{c}d$

$\bar{a}bc + a\bar{b}c$

$\bar{a}\bar{b}\bar{c}\bar{d} + \bar{a}b\bar{c}\bar{d} + \bar{a}\bar{b}c\bar{d} + \bar{a}b\bar{c}d + \bar{a}\bar{b}cd + \bar{a}b\bar{c}d + \bar{a}bcd + abcd$
 $\bar{a}b\bar{d} + \bar{a}b\bar{d} + \bar{a}\bar{c}d + \bar{a}bc + a\bar{b}c + abc$

$\bar{a}b\bar{d} + \bar{a}b\bar{d} + a\bar{c}d + abc$

$ac + ab + ac + ab + bc + bc$
 $a + a + b + b + c + c$



$$\bar{a}\bar{b}\bar{c} + \bar{a}bc + a\bar{b}\bar{c} + ab\bar{c} + abc$$

$$\bar{a}\bar{b} + a\bar{c} + ab$$

$$b + a\bar{c}$$

$$\begin{array}{cc} 010 & 110 \\ 011 & 111 \end{array}$$

$$b + a\bar{c}$$

$\bar{a}b$	00	01	11	10
c				
0		1	1	1
1		1	1	

$$\bar{a}b\bar{c} + \bar{a}b\bar{c} + \bar{a}bc + a\bar{b}\bar{c} + abc + a\bar{b}\bar{c}$$

$$\bar{a}\bar{b}$$

$$+ \bar{a}\bar{b} + a\bar{c}$$

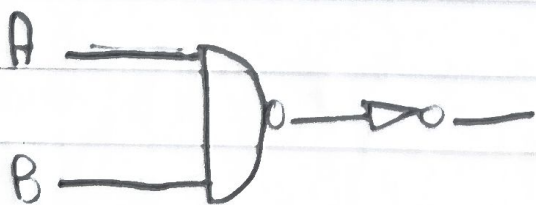
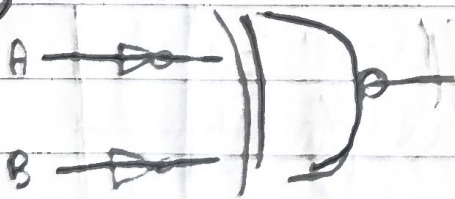
$\bar{a}b$	00	01	11	10
c				
0	1	1	1	1
1		1	1	

$$\bar{a}\bar{b} + \bar{a}\bar{b} + a\bar{c}$$

$$c + \bar{a}\bar{b} + a\bar{c}$$

$$01\bar{0} \ 01\bar{1} \ 10\bar{0} \ 10\bar{1}$$

7

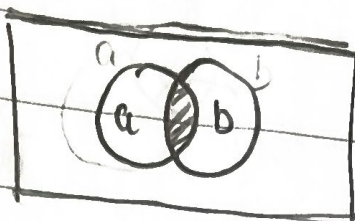


$$A \cdot B$$

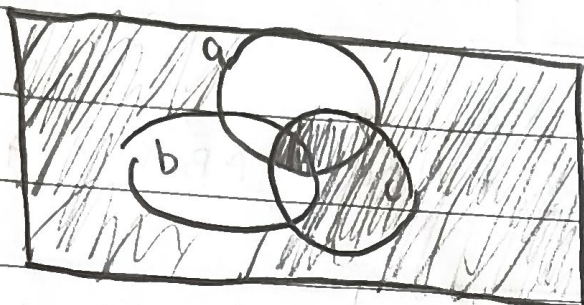
$$A + B$$

$$A \cdot B$$

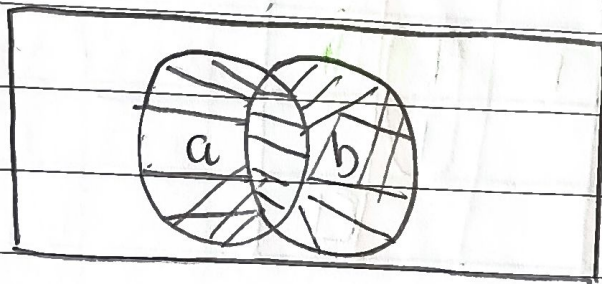
8 a) $A.B$



b) $\bar{a}\bar{b}\bar{c} + \bar{a}\bar{b}c + a\bar{b}c + abc$



d) $\bar{A}.B + \bar{A}\bar{B} + AB$



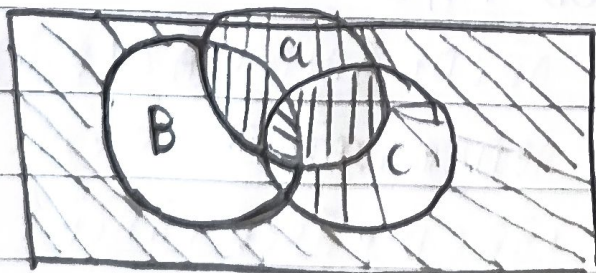
e) $ABC + \bar{A}B\bar{C} + AB\bar{C} + A\bar{B}\bar{C} + \bar{B} + A + AC$

$$ABC + \bar{A}B\bar{C} + AB\bar{C} + A\bar{B}\bar{C} + (A+\bar{A}).(\bar{C}+C).\bar{B} + A(B.\bar{B}).(\bar{C}+C) + AC(B+\bar{B})$$

$$ABC + AB\bar{C} + \cancel{A\bar{B}\bar{C}} + \cancel{A\bar{B}C} + AC\bar{B} + \cancel{A\bar{C}\bar{B}} + \bar{A}\bar{C}\bar{B} + \bar{A}\bar{C}B + \cancel{ABC} + \cancel{AB\bar{C}} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \cancel{ABC} + \cancel{A\bar{B}C}$$

$$ABC + AB\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}$$

$$\begin{matrix} ABC & + & AB\bar{C} & + & A\bar{B}\bar{C} & + & \bar{A}\bar{B}C & + & \bar{A}\bar{B}\bar{C} & + & \bar{A}\bar{B}C & + & \bar{A}\bar{B}\bar{C} \\ x & & x & & x & & x & & x & & x & & x \end{matrix}$$



9) a) $\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + ABC$

ab \ cd	00	01	11	10
00	1	1		
01			1	1
11				
10	1			

$\bar{a}\bar{b}$
 $\bar{a}b + bc$

b)

$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD + ABC\bar{D} + ABCD$

ab \ cd	00	01	11	10
00	1			1
01	1	1	1	1
11			1	
10	1			1

$0000 + 0001 + 0010 + 0011 + 0100 + 0101 + 0110 + 0111 + 1000 + 1001 + 1010 + 1011 + 1100 + 1101 + 1110 + 1111$
 $\bar{a}\bar{b}\bar{c} \quad \bar{c}d \quad \bar{b}c\bar{d} \quad abd$

$\bar{a}\bar{b}\bar{c} + \bar{c}d + \bar{b}c\bar{d} + abd$

c)

$\bar{B}\bar{D} + A + A\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}\bar{C}$

$(A+\bar{A})(\bar{B}+\bar{B})(\bar{C}+\bar{C})(\bar{D}+\bar{D}) + A(B+\bar{B})(C+\bar{C})(D+\bar{D}) + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}\bar{C}(B+\bar{B})(D+\bar{D})$
 $A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD + AB\bar{C}\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} + AB\bar{C}D + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D$
 $A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}\bar{D}$

$$9c) \quad \bar{B}\bar{D} + A + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}\bar{C}$$

$$\begin{aligned} & \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} \\ & \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} \end{aligned}$$

$$(A+\bar{A}) \cdot (C+\bar{C}) \cdot \bar{B}\bar{D}$$

$$\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D}$$

$$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D}$$

$$\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D}$$

$$\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D}$$

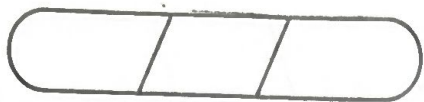
$$\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D}$$

$$\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D}$$

$$\begin{aligned} & \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} \\ & \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} \end{aligned}$$

$\bar{C}\bar{D} \backslash ab$	00	01	11	10
00	1		1	1
01			1	1
11	1		1	1
10	1		1	

$$\begin{aligned} & \begin{array}{cc|cc|cc|cc} 0000 & 1000 & 0011 & 0010 & 1100 & 1111 & 1101 & 1001 & ad \end{array} \\ & \begin{array}{cc|cc|cc|cc} \bar{b}\bar{c}\bar{d} & + & \bar{a}\bar{b}c & & 1101 & 1110 & + ab & 1111 & 1011 & + ad \end{array} \\ & \bar{b}\bar{c}\bar{d} + \bar{a}\bar{b}c + ab + ad \end{aligned}$$



10 b) b c d k

0 0 0 0	1	
0 0 0 1	0	1
0 0 1 0	X	
0 0 1 1	1	
0 1 0 0	1	
0 1 0 1	0	1
0 1 1 0	1	X
0 1 1 1	1	
1 0 0 0	1	
1 0 0 1	1	X
1 0 1 0	0	1
1 0 1 1	1	
1 1 0 0	1	
1 1 0 1	X	
1 1 1 0	0	1
1 1 1 1	1	

a) a b c d k

0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	X
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	X
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Figura 7.

$$(\bar{A} + B + \bar{C} \cdot \bar{D}) + ((A + \bar{C}) \cdot B)$$

$$A \oplus B$$

$$\bar{A}B + A\bar{B}$$

$$(\bar{A} + B + \bar{C} + D) \cdot (\bar{A}B + B\bar{C}) \cdot \bar{A}B + A\bar{B}$$

$$(\bar{A} + B + \bar{C} + D) \cdot ((\bar{A} + \bar{B}) + (\bar{B} + C)) \cdot \bar{A}B + A\bar{B}$$

$$(\bar{A} + B + \bar{C} + D) \cdot (\bar{A}\bar{B} + \bar{A}C + \bar{B} + \bar{B}C) \cdot \bar{A}B + A\bar{B}$$

$$\bar{A}\bar{B}\bar{A}B + \bar{A}\bar{B}/A\bar{B} + \bar{A}C\bar{A}B + \bar{A}/A\bar{B} + \bar{A}\bar{B}\bar{B} + A\bar{B}\bar{B} + \bar{A}\bar{B}C + \bar{B}/A\bar{B}$$

$$(\bar{A} + B + \bar{C} + D) \cdot (\bar{A}BC + A\bar{B})$$

$$\bar{A}BC + \bar{A}\bar{B}C + A\bar{B}\bar{C} + \bar{A}B\bar{C}D + A\bar{B}D$$

$$\bar{A}BC + \bar{A}\bar{B}C + A\bar{B}D$$

Figura 8

$$((\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C) + (\bar{A}BC + A\bar{B}C)) + (ABC)$$

$$\bar{A}\bar{B} + \bar{A}BC + A\bar{B}C + ABC$$

ab \ c	00	01	11	10
0	1			
1	1	1	1	1

001 001 111 101
 $C + \bar{a}b$
 000 00x

Figura 9

$$AC + B + D + \overline{ACD} \cdot C$$

$$AC \cdot \bar{B} \cdot \bar{D} + (\bar{A}\bar{C}\bar{D} \cdot C)$$

$$(\bar{A}\bar{C}\bar{D}) + \bar{A}C + C\bar{D}$$

$$\bar{A}C + C\bar{D}$$

ab \ c	00	01	11	10
00				
01				
11		1		1
10	1	1	1	1

$$\bar{A}C(B + \bar{B}) \cdot (D + \bar{D})$$

$$\bar{A}BC$$

$$\bar{C}\bar{D} (B + \bar{B}) (A + \bar{A})$$

$$\bar{A}BCD + \bar{A}BC\bar{D}$$

$$ABC\bar{D} + A\bar{B}C\bar{D}$$

$$\bar{A}BCD + \bar{A}\bar{B}C\bar{D}$$

$$\bar{A}BC\bar{D} + \bar{A}\bar{B}C\bar{D}$$